



65W USB-C Power Delivery Charger

Electrical and Computer Engineering Program

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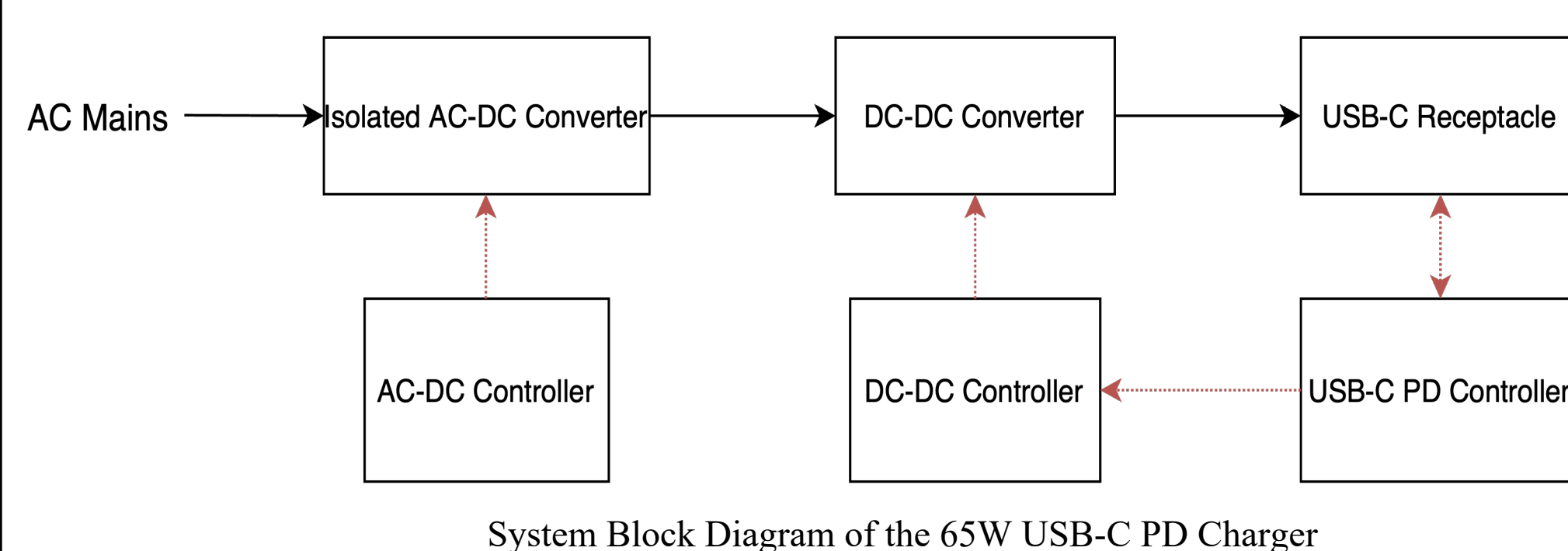


Abstract

This project details the design of a compact 65W USB-C Power Delivery (PD) charger that meets the fast-charging requirements of modern devices. Key objectives included minimizing power losses, reducing costs, and decreasing size while ensuring reliable USB-C PD operation. The resulting charger balances efficiency and size, demonstrating strong performance.

Introduction

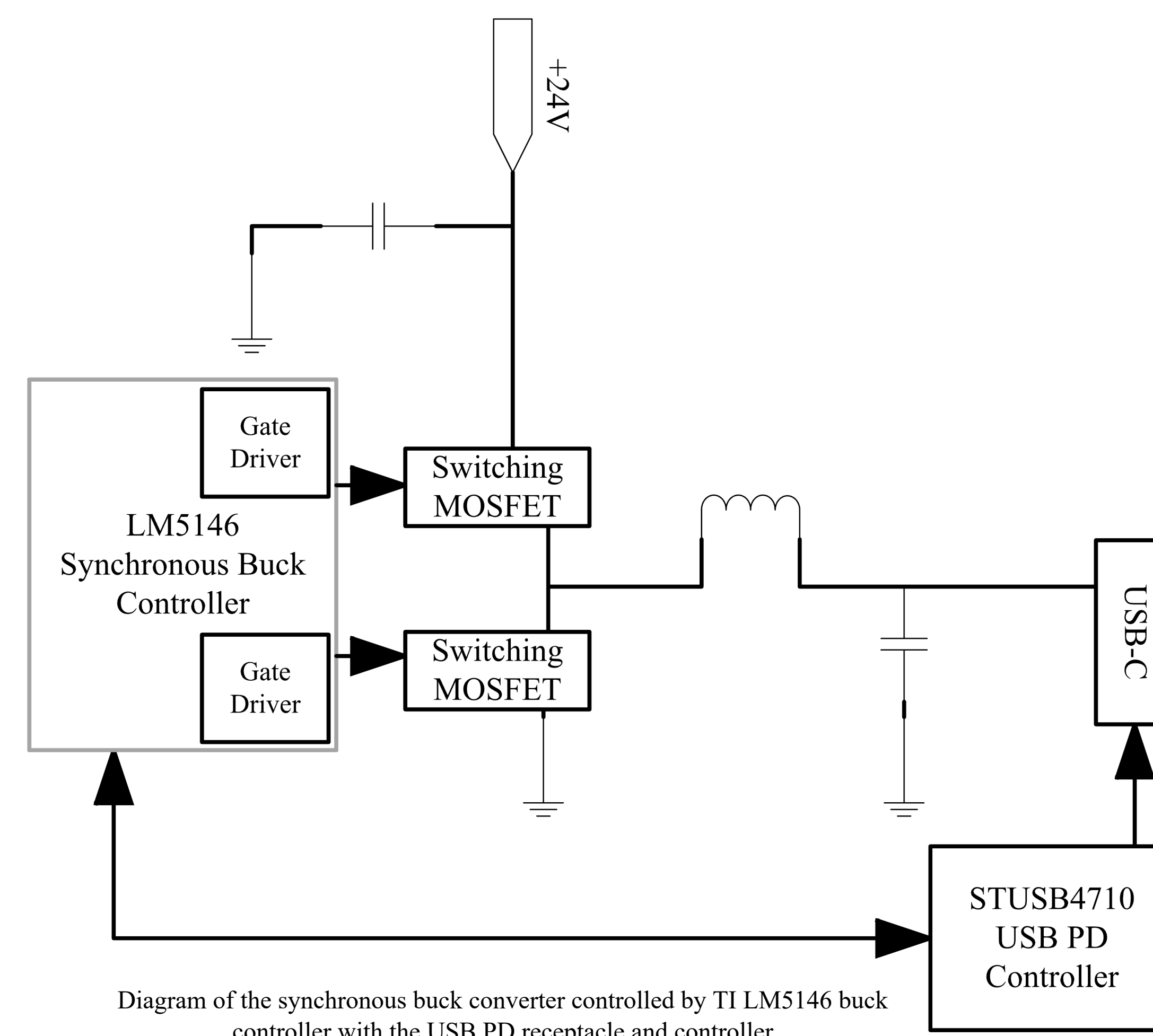
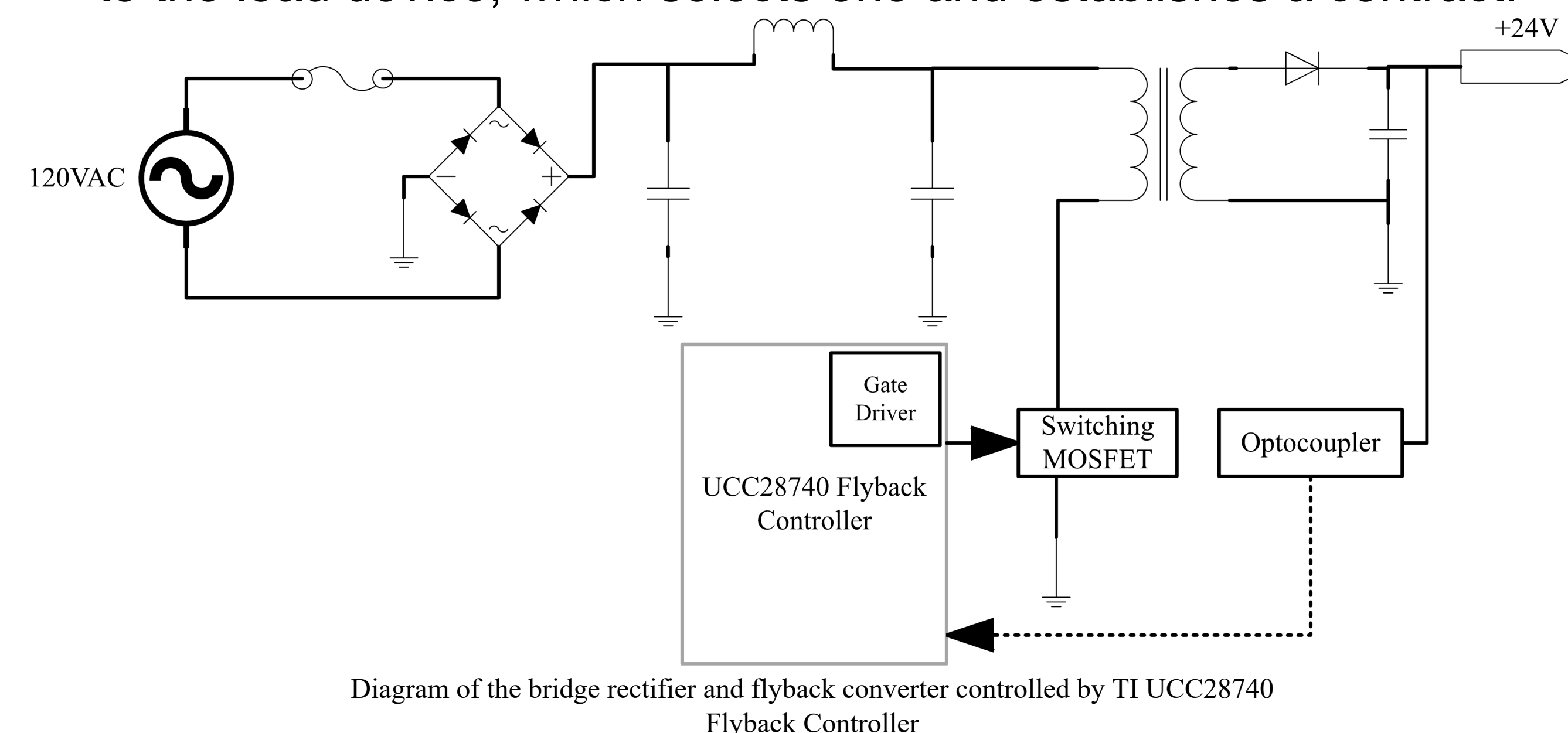
USB Power Delivery (PD) is a system that enables devices to negotiate their power requirements over a USB connection. High-frequency switch-mode power supplies enable the use of smaller components, resulting in a more compact overall power supply. The objective of this project was to create a universal charger that powers both small and large devices, ensuring compatibility with a wide range of devices.



Materials & Methods

The charger employs a two-stage design consisting of a quasi-resonant flyback stage followed by a synchronous buck stage. This configuration reduces topology complexity and performance in power conversion.

- The flyback stage with a bridge rectifier converts $120V_{rms}$ into $24V_{DC}$.
- The synchronous buck converter reduces the input voltage from 24V to a range between 5V and 20V, depending on the load.
- The USB PD controller communicates Power Data Objects (PDOs) to the load device, which selects one and establishes a contract.

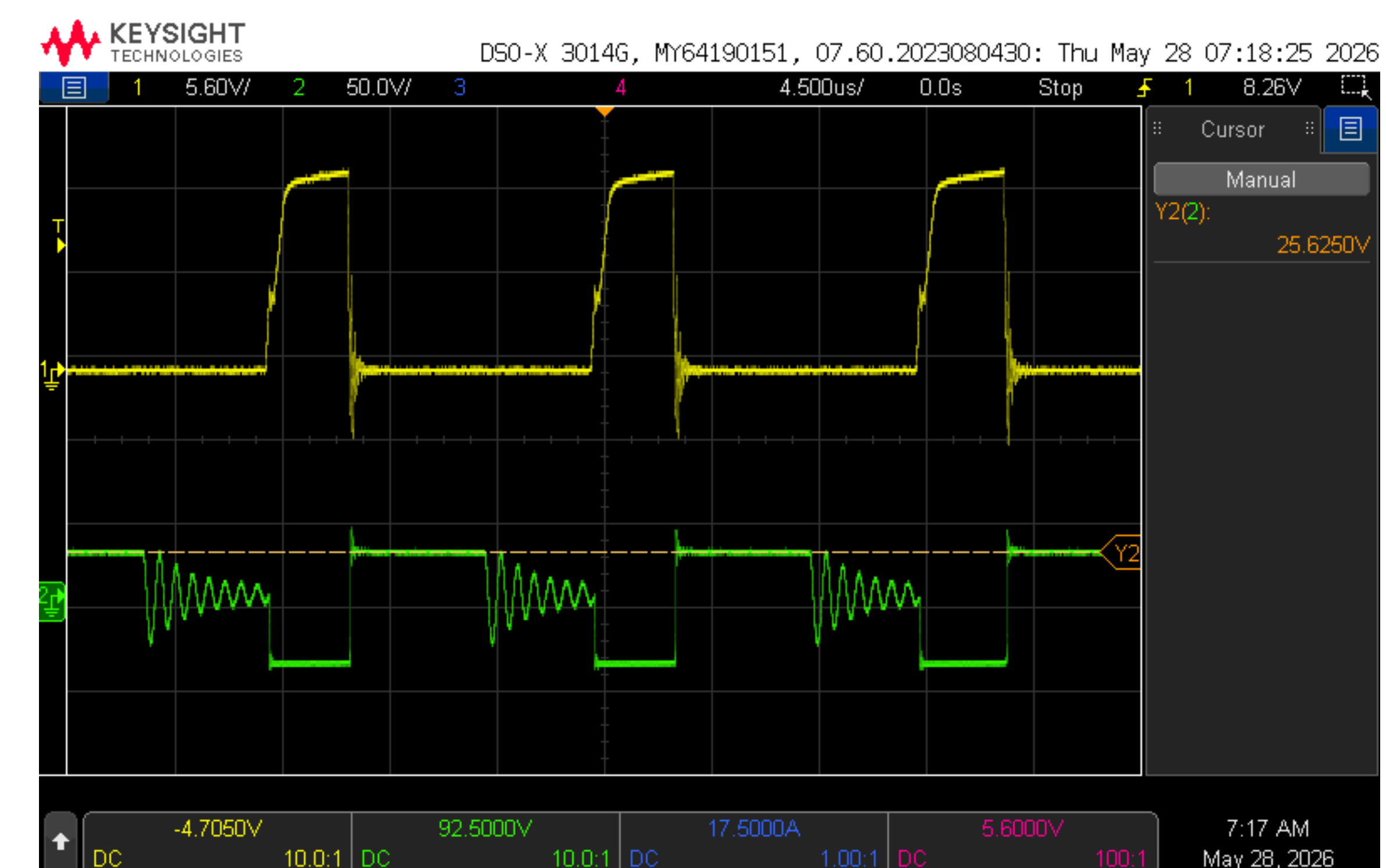


Results

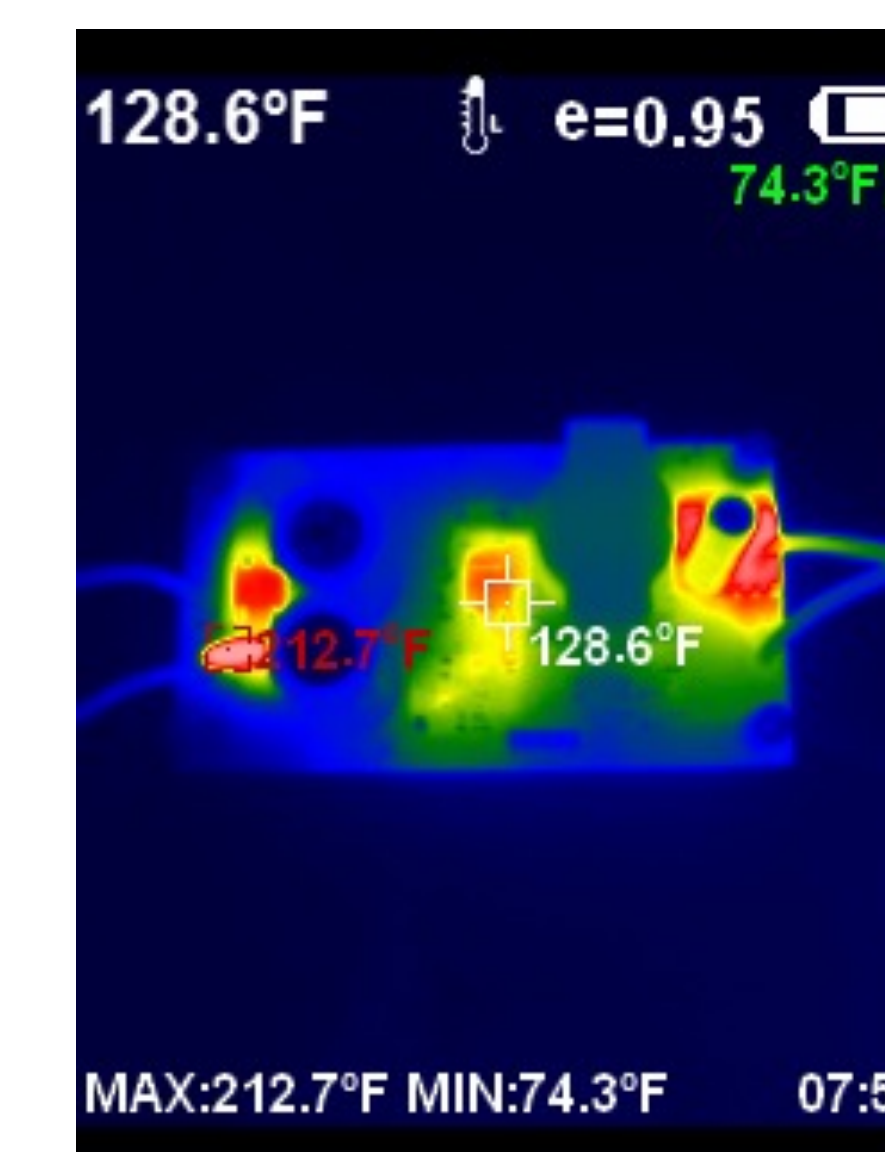
The flyback and buck stages were tested independently to verify proper functionality before integration into a cohesive system.

- Buck stage 5V – 20V output; with 24V input.
- Flyback stage 24V output; with $120V_{rms}$ input.
- Charger outputs 1W – 58W
- USB PDOs 5V/9V/12V/15V/20V

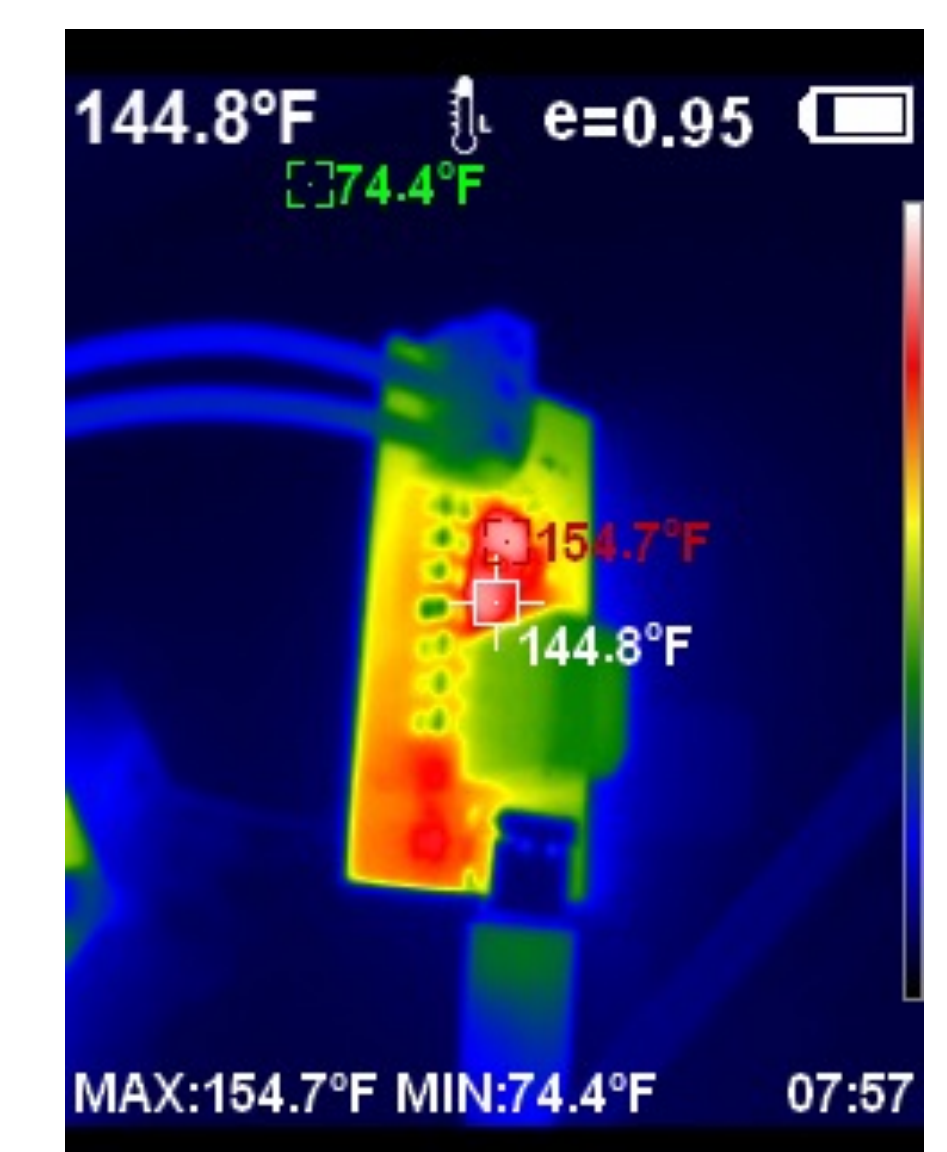
Below is the charger, tested at $120V_{RMS}$ with an AC power supply and loaded with an electronic load in constant-current mode @ 2.9A.



The oscilloscope probe of the flyback stage shows 25.5V output from the secondary winding. The flyback MOSFET gate waveform is in yellow, and the secondary winding waveform is in green.



Thermal imaging of the flyback converter loaded to 24V @ 2.7A. The highest temperature seen is 212.7°F



Thermal imaging of the buck converter loaded to 20V @ 2.9A. The highest temperature seen is 154.7°F